White Paper 2 — Druggability annotation via knowledge bases

Pipeline step: Druggability annotation via knowledge bases **Inputs:** candidate gene list (168 genes, STAT3-anchored) **Outputs:** `druggability_annotation.csv`, `fig_druggability_landscape.png`

1. Purpose

A differentially expressed gene is a biological hypothesis; a *druggable* differentially expressed gene is a therapeutic hypothesis. This step annotates each candidate protein with what is known about its amenability to small-molecule intervention — drawing on two curated knowledge bases rather than inferring druggability from scratch — so that the leads chosen for structure-based work are those with a realistic path to a molecule.

2. The concept of druggability

The "druggable genome" framing of Hopkins and Groom established that only a subset of human proteins present binding sites with the physicochemical properties needed to bind drug-like small molecules, and that membership in privileged protein families (enzymes, GPCRs, nuclear receptors, ion channels) is a strong prior for tractability [1]. Druggability is therefore not a single number but a composite of pocket chemistry, protein family, and precedent — which is exactly what the knowledge bases encode.

3. Open Targets: tractability and target class

The **Open Targets Platform** integrates genetic, genomic, and pharmacological evidence to support systematic target identification and prioritisation [2]. For each candidate we pulled:

- **Small-molecule tractability bucket** — Open Targets' graded assessment running from "Approved drug" (a marketed small molecule exists) through "Advanced clinical", "Phase 1", "Structure with ligand / druggable pocket", down to "Discovery precedence" and "Unknown". This bucket is the single most informative druggability descriptor because it encodes real-world precedent.
- **Target class** — the protein family (enzyme, transcription factor, membrane receptor, secreted), which sets the prior on pocket type and available chemistry.
- **Known drugs and clinical candidates** — the count of molecules already in development against the target.

Across the 154 mapped candidates, the tractability distribution was: Approved drug 8, Advanced clinical 14, Phase 1 1, Druggable pocket 5, Discovery precedence 40, Unknown 86 — i.e. roughly a third of psoriasis DEGs have at least discovery-level small-molecule precedent.

4. ChEMBL: quantitative chemical precedent

Where Open Targets gives a graded label, **ChEMBL** gives the underlying medicinal-chemistry evidence: a manually curated database of bioactive molecules with measured potencies against protein targets [3]. For each candidate mapped to a single-protein ChEMBL target we counted **potent inhibitors**, defined by the standardized **pChEMBL** value — a harmonized $-\log_{10}$ activity (IC_{50}/K_i/EC_{50}) that makes potencies comparable across assays [4]. A pChEMBL ≥ 7 threshold (≤ 100 nM) counts genuinely potent chemical matter.

The inhibitor counts sharply stratified the candidates: RORC (RORγt) 9,766 potent inhibitors, JAK3 6,069, versus STAT3 just 112 — a quantitative reflection of how much tractable chemistry each pocket has attracted.

5. Disease-axis relevance guards against off-target tractability

Raw tractability can mislead: a gene may be highly druggable because of an *unrelated* drug program (a neurotransmitter receptor, a renin inhibitor) that has nothing to do with psoriasis. We therefore annotated each candidate with its **psoriasis disease axis** — Th17/IL-17, JAK-STAT, NF-κB/TNF, innate/antimicrobial, chemokine/leukocyte, tryptophan/immunometabolism, eicosanoid/protease, and proliferation. This axis layer is what keeps the downstream lead selection mechanistically coherent rather than merely chemically convenient. The biological grounding for the dominant axes is well established: RORγt as the master transcription factor of the IL-17-producing T-helper lineage [5], the IL-17 cytokine family as central effectors of tissue inflammation [6], STAT3 as the node linking activated keratinocytes to the immune infiltrate in psoriasis [7], and JAK3 as an essential kinase in cytokine-receptor signaling whose loss causes immunodeficiency [8].

6. Deliverables

- `druggability_annotation.csv` — 154 candidates with tractability bucket, target class, known-drug count, potent-inhibitor count, and disease axis.
- `fig_druggability_landscape.png` — the druggability landscape, positioning each candidate by DE evidence versus chemical/structural tractability.

7. Caveats

- Knowledge-base annotations are a snapshot; tractability buckets and inhibitor counts evolve as new chemistry is deposited.
- 14 candidates (lincRNAs, uncharacterized clones) did not map to a protein target and carry no druggability annotation.

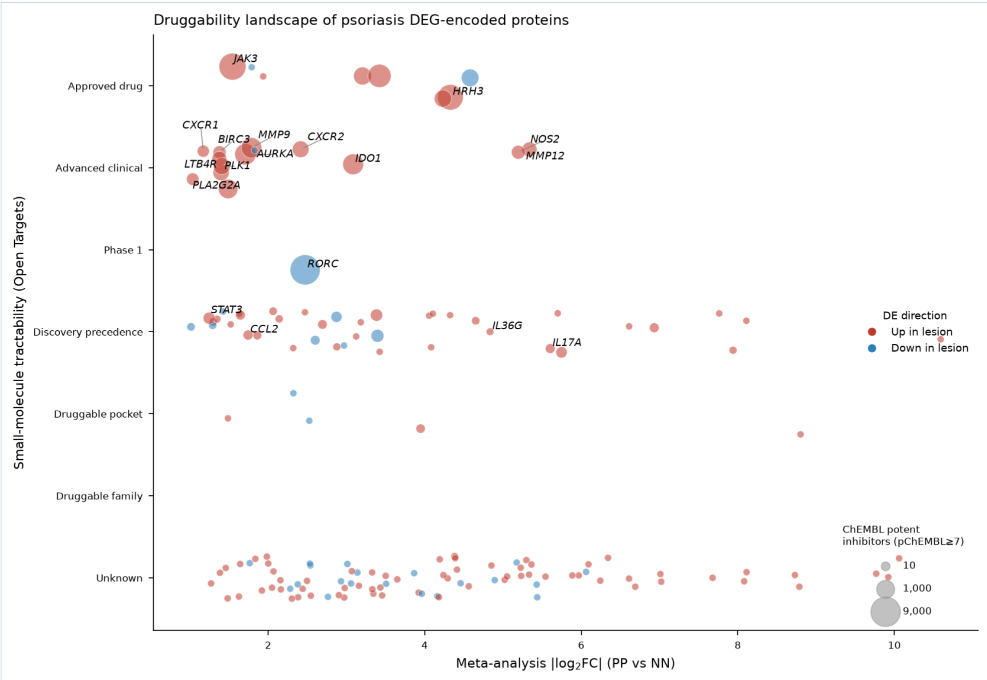

Figure: key output of this pipeline step.

References

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